

Introduction to Contract Theory and Auctions
More about revenues & Revenue Equivalence Theorem

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Expected Payment once again...

- Recall that in both auction formats, FPA and SPA, or for short I and II respectively, we have that the **interim expected payment** for each individual buyer is given by

$$m^A(v) = G(v) \times E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v]$$

where $A = I, II$.

- Thus, the ex-ante payment for each individual buyer is given by

$$E[m^A(V)] = \int_0^{\bar{v}} m^A(v) f(v) dv$$

- Note that the auctioneer's expected revenue in each auction format, R^A , is given by

$$E[R^A] = n E[m^A(V)] = n \int_0^{\bar{v}} y(1 - F(y)) g_1^{(n-1)}(y) dy$$

The Revenue Equivalence Theorem

- Consider a SIPV sealed-bid auction described by (q, p)
 - Recall that $q : [0, \infty)^n \rightarrow \Delta(N)$, gives the winning probability in the auction as a function of the vector of bids.
 - $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ gives the payment vector for each player as a function of the identity of the winner and the vector of bids.
- Let $\beta : \mathcal{V} \rightarrow [0, \infty)$ be a monotonically increasing strategy.
- Denote by $m_i(v_i) = m(q, p, \beta; v_i)$ the expected payment that buyer i with private value v_i pays in an auction method (q, p) when all the buyers implement β :

$$m_i(v_i) = \int_{v_{-i} \in \mathcal{V}^{n-1}} \sum_{i_* \in N} q_{i_*}(\beta(v_1), \dots, \beta(v_n)) p_i(i_*, \beta(v_1), \dots, \beta(v_n)) dF_{-i}(v_{-i})$$

Theorem 6.1

Let β be a symmetric and monotonically increasing equilibrium in a sealed-bid auction with independent private values satisfying the following properties:

- 1 the winner of the auction is the buyer with the highest private value.
- 2 the expected payment by a buyer with private value 0 is 0.

Then

$$m_i(v_i) = G_1^{(n-1)}(v_i) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v_i].$$

- Property (1) is called the **efficiency condition**: an efficient auction is one in which, in equilibrium, the object is allocated to the buyer who most highly values it.
- If the seller is willing to sell the object at any nonnegative price, and since the private values of the buyers are nonnegative, an efficient auction **allocates the object with probability 1**.
- What is important about theorem 7.3: the right hand side shows that the expected payment of any bidder **does not depend on the auction method!** (Caveat: assuming the SIPV assumptions and risk-neutrality of the buyers.)

Corollary 6.2 (The Revenue Equivalence Theorem)

In a symmetric-sealed bid auction with independent private values, let β be a symmetric and monotonically increasing equilibrium satisfying the following properties (1) and (2) from theorem 7.3. Then the seller's expected revenue is

$$\pi = n \int_{\mathcal{V}} m_i(v) f(v) dv$$

where

$$m_i(v) = G_1^{(n-1)}(v) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v].$$

We now use the RET to find symmetric equilibria in various auctions.

Theorem 6.3

Let β be a symmetric, monotonically increasing equilibrium strategy, satisfying $\beta(0) = 0$ in a symmetric FPA with independent private values. Then

$$\beta(v) = \mathbb{E}[Y_1^{(n-1)} \mid Y_1^{(n-1)} \leq v]$$

This theorem complements Thm. 5.10, since it shows that the equilibrium we found in that theorem is the unique such symmetric equilibrium in the FPA.

Equilibrium Strategy in an All-Pay Auction

Suppose that β^{all} is a symmetric, monotonically increasing equilibrium strategy for the all-pay SIPV auction. Suppose that $\beta^{\text{all}}(0) = 0$. We can use the RET to derive the shape of β^{all} .

In an all-pay auction

$$m_i(v; \text{all}) = \beta^{\text{all}}(v).$$

The RET yields

$$m_i(v; \text{all}) = \int_0^v s g_1^{(n-1)}(s) ds,$$

which implies that

$$\beta^{\text{all}}(v) = \int_0^v s g_1^{(n-1)}(s) ds.$$

Equilibrium Strategy in an All-Pay Auction

It remains to check that the above is an equilibrium. Suppose that all bidders except i follow β^{all} . If i bids $\beta^{\text{all}}(z)$ for some value $z \in \mathcal{V}$ when her actual value is v , then her expected payoff is

$$G_1^{(n-1)}(z)v - \beta^{\text{all}}(z) = G_1^{(n-1)}(z)v - \int_0^z s g_1^{(n-1)}(s) ds.$$

Integrating by parts yields:

$$G_1^{(n-1)}(z)(v - z) - \int_0^z G_1^{(n-1)}(s) ds.$$

The first-order condition is:

$$g_1^{(n-1)}(z)(v - z) = 0 \Rightarrow z = v.$$

- Although we know that $E[R^I] = E[R^{II}]$, it is not the case that R^I and R^{II} are the same random variable.

Example 6.4

Recall example 6.2, two bidders with $V_1, V_2 \sim U[0, 1]$ i.i.d. and $\beta^I(v) = \frac{v}{2}$ and $\beta^{II}(v) = v$. Then, by direct calculation:

$$R^I = \max\{\beta^I(V_1), \beta^I(V_2)\} = \beta^I(\max\{V_1, V_2\}) = \beta^I(Y_1^{(2)}) = \frac{Y_1^{(2)}}{2}$$

$$R^{II} = \min\{V_1, V_2\} = Y_2^{(2)}$$

- Let F_I be the distribution of R^I and F_{II} be the distribution of R^{II} .
- Example 7.1 suggests that F_{II} is a **mean-preserving spread** of F_I .
- This means that from the perspective of the auctioneer, the SPA is **riskier** than the FPA.
- Indeed, any **risk-averse auctioneer** would strictly prefer a first-price auction versus a second-price auction (assuming the bidders are risk-neutral).

Proposition 6.5

*In the SIPV auction model, the distribution of equilibrium payments in a second-price auction is a mean-preserving spread of the distribution of equilibrium payments in a first-price auction. That is, F_I **second-order stochastically dominates** F_{II} .*

Proof.

